1 $\left(-\frac{1}{4}\right)^2$ or $\frac{1}{16}$ **[M1]**

$4 \times \frac{1}{16}$ or $\frac{1}{4}$ **[M1]**

$-3\frac{1}{4}$ or $-\frac{13}{4}$ or -3.25 **[A1]**

[Total Marks: 3]

2 **(a)** 9 **[B1]**

(b) 25 **[B1]**

(c) 27 **[B2]**

If 0 scored, M1 for -4 or -3 seen **[M1]**

[Total Marks: 4]

3 **(a)** 34 **[B2]**

M1 for $-2 \times -3 = 6$ or $12 - (-6) + 16$ or $18 + 16$ seen

(b) 1 **[B2]**

M1 for $\frac{1}{6} \times \frac{3}{2} = \frac{3}{12}$ or $\frac{1}{4}$ seen

[Total Marks: 4]

4 **(a)** Evaluates multiplication and division correctly to get $+12$ and -5 **[M1]**

19 **[A1]**

(b) Correct division seen (e.g. $\frac{2}{5} \times \frac{7}{4}$ or $\frac{14}{20}$ or $\frac{7}{10}$) **[M1]**

$\frac{3}{5}$ **[A1]**

[Total Marks: 4]

5 **(a)** $15 - 3 \times 6$ or $15 - 18$ **[M1]**

-3 **[A1]**

(b) $\frac{9}{4} \times \frac{8}{3}$ **[M1]**

6 **[A1]**

$4\frac{1}{2}$ or $\frac{9}{2}$ **[A1]**

[Total Marks: 5]

6 **(a)** 28 **[B2]**

If 0 scored, allow [M1] for correctly evaluating the bracket to -4 or for showing $12-(-16)$

(b) 30 **[B2]**

If 0 scored, allow [M1] for an equivalent fraction such as $\frac{120}{4}$ or $\frac{12}{0.4}$

(c) 3.5 **[B1]**

[Total Marks: 5]

7 (a) $1\frac{13}{30}$ **[B3]**

If 0 scored, allow [M1] for correctly evaluating the multiplication as $\frac{18}{30}$ or $\frac{3}{5}$

Allow a further [M1] for finding a correct common denominator to add the fractions, e.g., $\frac{25}{30}+\frac{18}{30}$ or giving the improper fraction $\frac{43}{30}$

(b) 6 **[B3]**

If 0 scored, allow [M1] for correctly evaluating the bracket as $\frac{3}{8}$

Allow a further [M1] for converting to an improper fraction and correctly multiplying, e.g., $\frac{9}{4}\times\frac{8}{3}$

[Total Marks: 6]

8 (a) $\frac{8}{6}-\frac{5}{6}$ or $\frac{1}{2}$ **[M1]**

$\frac{12}{5}\div$ their $\frac{1}{2}$ **[M1]**

$\frac{12}{5}\times\frac{2}{1}$ or equivalent multiplication **[M1]**

$4\frac{4}{5}$ **[A1]**

(b) 0.54 or $\frac{45\times 12}{600}$ or equivalent **[M1]**

$54\div 6$ or $0.54\div 0.06$ **[M1]**

9 **[A1]**

[Total Marks: 7]

9 Denominator: $2-0.4=1.6$ or $\frac{8}{5}$ **[B1]**

Right hand side: $-1\frac{1}{4}=-\frac{5}{4}$ **[B1]**

$x+\frac{1}{3}=-\frac{5}{4}\times\frac{8}{5}$ **[M1]**

$x+\frac{1}{3}=-2$ **[A1]**

$x = -\frac{7}{3}$ or $-2\frac{1}{3}$ **[A1]**

[Total Marks: 5]

10 Numerator multiplication: $\frac{7}{3} \times \frac{9}{7} = 3$ **[M1]**

Evaluates root as $\frac{5}{3}$ and cube as $\frac{1}{8}$ **[M1]**

Denominator division: $\frac{1}{8} \times \frac{16}{3} = \frac{2}{3}$ **[M1]**

Numerator subtraction: $3 - \frac{5}{3} = \frac{4}{3}$ **[M1]**

2 **[A1]**

[Total Marks: 5]

11 Converts mixed numbers to improper fractions and multiplies correctly: $\frac{9}{4} \times \frac{4}{3} = 3$ (or $\frac{12}{4}$ or $\frac{36}{12}$) **[M1]**

Evaluates the numerator correctly: $3\frac{2}{3} + 3 = 6\frac{2}{3}$ (or $\frac{20}{3}$) **[M1]**

Evaluates the denominator correctly using a common denominator: $\frac{27}{6} - \frac{11}{6} = \frac{16}{6} = \frac{8}{3}$ **[M1]**

Sets up the correct division for the final step: $\frac{20}{3} \div \frac{8}{3} = \frac{20}{3} \times \frac{3}{8}$ (or $\frac{20}{8}$) **[M1]**

$\frac{5}{2}$ or $2\frac{1}{2}$ **[A1]**

[Total Marks: 5]

12 (a) $6\frac{2}{5}$ **[B4]**

M1 for changing mixed numbers to improper fractions, e.g., $\frac{5}{4}$ or $\frac{8}{3}$

M1 for writing fractions with a common denominator to subtract, e.g., $\frac{15}{12} - \frac{10}{12}$

M1 for $\frac{8}{3} \times \frac{12}{5}$ or $\frac{32}{5}$

(b) $38\frac{1}{3}$ **[B4]**

M1 for $83\frac{1}{3} \div 12\frac{1}{2}$

M1 for $\frac{250}{3} \times \frac{2}{25}$ or width $= \frac{20}{3}$ $(6\frac{2}{3})$

M1 for $2 \times \left(12\frac{1}{2} + \frac{20}{3}\right)$ or $2 \times \left(\frac{75}{6} + \frac{40}{6}\right)$

[Total Marks: 8]

13 **(a)** Finds the temperature at 14:00 ($-4.5 + 11.2 = 6.7°C$) or sets up the full expression $(-4.5 + 11.2 - 8.1)$ **[M1]**

-1.4 **[A1]**

(b) Evaluates numerator to 19.63... or denominator to 5.115 **[M1]**

3.84 **[A1]**

[Total Marks: 4]

14 Net rate of fill $= 0.75 - 0.45 = 0.3$ litres/min oe **[M1]**

Total time $= \dfrac{85.5}{0.3} = 285$ minutes **[M1]**

$285 \div 60 = 4$ with remainder 45 **[M1]**

4 hours 45 minutes **[A1]**

[Total Marks: 4]

15 **(a)** 37.5 **[B2]**

M1 for 15×2.8 or 42 seen

(b) 23 **[B3]**

M1 for $37.5 - 1.2m = 9.9$ or $37.5 - 9.9$ or 27.6 seen

M1 for $27.6 \div 1.2$

[Total Marks: 5]

16 **(a)** $12.5 - (3.2 \times 8)$ or $12.5 - 25.6$ **[M1]**

-13.1 **[A1]**

(b) $(-4.5 \times -2) - 18.3$ **[M1]**

Final temperature of Sample B $= -9.3$ **[A1]**

Difference $= |-9.3 - (-13.1)|$ or $13.1 - 9.3$ **[M1]**

3.8 **[A1]**

[Total Marks: 6]

17 **(a)** $\dfrac{2}{5} \times 1000$ or 400 **[M1]**

18.5×20 or 370 **[M1]**

770 **[A1]**

(b) Target volume $= 500$ or volume to lose $= 770 - 500 = 270$ **[M1]**

$270 \div 4.5$ **[M1]**

60 **[A1]**

(c) Volume to fill $= 1000 - 500 = 500$ **[M1]**

1 hour 20 minutes = 80 minutes **[M1]**

$500 \div 80$ leading to 6.25 **[A1]**

[Total Marks: 9]

18 (a) Number of bottles = 360 **[B1]**

Volume left over = 0.5 **[B2]**

If 0 scored for the volume left over, allow [M1] for $450.5 - (360 \times 1.25)$

(b) 4 hours 54 minutes **[B4]**

If 0 scored, allow [M1] for calculating the volume of the pool: $24.5 \times 12 \times 1.8 = 529.2$ m^3

Allow a further [M1] for converting the volume to litres: $529.2 \times 1000 = 529200$ litres

Allow a further [M1] for calculating total minutes: $\dfrac{529200}{1800} = 294$ minutes

(c) -8.5 **[B3]**

If 0 scored, allow [M1] for correctly interpreting 6 hours 20 minutes as a fraction or decimal of hours, e.g., $6\dfrac{1}{3}$ hours or $\dfrac{19}{3}$ hours

Allow a further [M1] for calculating the total temperature rise: $1.5 \times \dfrac{19}{3} = 9.5$

[Total Marks: 10]